CptS 484: Software Requirements

WRS Evolution

Requirements Elicitation

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Revision History

Date	Version	Changes	Editor
9/17/25	1.0	Initial Draft	Joshua Allard, Emily Porter, Noah Schmalenberger
9/18/25	1.1	Updated Functional Requirements	Joshua Allard
10/1/25	1.2	Updated Preliminary Domain and Domain Issues	Nicole Fuller
10/1/25	1.3	Updated Preliminary Stakeholder and Stakeholder Issues	Emily Porter
10/1/25	1.4	Updated section 3.3	Noah Schmaleberger
10/7/25	1.5	Updated section 4.1	Noah Schmaleberger
10/7/25	1.6	Updated Section 2.4	Alena Watts
10/8/25	1.7	Updated Section 3.4	Alena Watts
10/11/25	1.8	Updated Traceability Matrix	Joshua Allard
10/11/25	1.9	Updated Prototype Interface Mockups and User Manual	Nicole Fuller
10/11/25	1.10	Added Section 1	Emily Porter
10/12/25	1.11	Updated Meeting schedule	Alena Watts
12/6/25	2.0	Update Section 1, Preliminary Stakeholder, and Stakeholder Issues with changes to preliminary definition	Emily Porter

12/7/25	2.1	Added goal KAOS models	Emily Porter
12/7/25	2.2	Updated Section 4.1 to reflect phase 2 changes to the preliminary definition	Noah Schmalenberger

[1] Introduction

1.1. Purpose

The purpose of this project is to develop a smartphone application that assists visually impaired people with indoor navigation. Theia, the smartphone application, will aid users in navigating from place to place within an indoor environment. Each building has its own layout and its own obstacles. Theia will adapt to obstacles (furniture, strange layouts, etc.) to ensure the safe transportation of its users. Additionally, to ensure that each user receives the most beneficial experience, Theia is configurable to each visually impaired user's needs. Ultimately, Theia's purpose is to provide a safe and easy way for visually impaired people to navigate indoor environments.

1.2. Scope

The Theia application will be a cross-platform mobile application, accessible on multiple types of smartphones, whether that be Apple, Samsung, or Android. It will use existing hardware components of a smartphone, such as cameras, microphones, and speakers. It will maximize utilization of sensors to be efficient with resources and ensure the best experience for users, incorporating fall detectors to notify emergency services if an incident occurs.

Using sensor data and spatial mapping, Theia will be able to assist its users in traversing whatever indoor environment the user is in. It will provide step-by-step instructions to effectively and safely direct users to their destination. This includes moving to a different floor, between buildings, and between rooms. From sensor data analysis, if there is an obstacle, Theia will be able to detect it and help users avoid or navigate through it. Overall, our application will be technically feasible as it uses existing hardware and a clear plan of implementation for software.

1.3. Objectives and Success Criteria

The objectives and success criteria that the:

- Adaptable to any indoor environment
- Communicate instructions clearly and effectively to visually impaired user
- Successfully direct visually impaired user to goal destination
- Works on multiple types of smartphones
- Maximizes utilization of sensors to increase efficiency
- Option to contact emergency services if needed
- Be easy to use
- Be configurable to each user's disability
- Protect user data to comply with HIPAA

1.4. Definitions, Acronyms, and Abbreviations

Term	Definition
HIPAA	Health Insurance Portability and Accountability Act
Visually impaired person	Any person with a visual disability, including being blind.
WRS	Website Requirements Specification

1.5. Overview

In Section 2, the preliminary domain, stakeholders, functional requirements, and nonfunctional requirements were identified. From there, in section 3, the issues with the domain, stakeholders, functional requirements, and nonfunction requirements were found and described. Each issue had different options for paths forward, and the chosen option was picked.
Section 4 is split into a couple of different parts for W and RS. The W section covers the type of problems that the application will cover and the corresponding goals. From there, in Section 4.1.3, improved understanding of the domain, stakeholders, functional objectives, and non-functional objects are recognized and described. In the RS part of the WRS, the types of requirements and their implementations are discussed.
Section 5 discusses the preliminary prototype, and Section 6 prototype interface mock-ups are provided. In Section 7, the user manual is implemented. Within Section 8, the traceability for functional and non-function requirements can be found.

[2] Preliminary Definition

2.1. Preliminary Domain

PD_ID	Preliminary Domain Description
PD1	Indoor navigation complexity
PD2	Obstacle detection ambiguity

2.2. Preliminary Stakeholder

PD_ID	Preliminary Stakeholder Description

PS1	Blind or visually impaired people who need to navigate indoors
PS2	Caretaker, someone who aides the blind person in configuring the application
PS3	Accessibility department staff who may upload indoor maps/layouts

2.3. Preliminary Functional Requirements

P FR_ ID	Preliminary FR Description
PFR1	Accept a location from the user to navigate to using voice commands or another form of user interface
PFR2	User can confirm destinations that have been suggested or selected by the user
PFR3	Figure out multiple routes to the users confirmed destination and inform them of the options as well as describe each route including time, distance, and potential obstacles.
PFR4	While navigating inform the user of their distance to the next direction
PFR5	While navigating, when reaching a stop or turn inform the user and tell them where to go
PFR6	Detect obstacles using sensors and tell user how to avoid them
PFR7	Place emergency calls and messages when a fall occurs or the system is lost
PFR8	Predict what the user's actions would be in advance using their location habits and schedule, suggesting their next destination and accepting the user's choice.
PFR9	User can create new emergency contacts for the system to use, including setting priority for their contacts
PFR10	User can customize settings such as volume, route update frequency, and language.

2.4.Preliminary Non-Functional Requirements

PNFR_ ID	Preliminary NFR Description
PNFR1	The system shall be secure
PNFR2	The system shall be usable
PNFR3	Generating desired sentences and representing them pictorially as well as associating with a sound/voice.
PNFR4	The system has a response time of about 2-3 seconds for user input under normal operating commands/conditions
PNFR5	System should be available daily, allowing for multi-user access, ensuring especially high availability during navigation and emergency situations
PNFR6	The system should securely save and encrypt users' personal data, such as their location history and username/email address. The system must also follow privacy protection rules and regulations.
PNFR7	The system should support the switching of multiple languages via settings, with clear voice communication and understanding of voice prompts for every language
PNFR8	The system should be reliable in all areas whether there is low lighting, crowds, or loud noise conditions, should also support when the user reports new obstacles in real time
PNFR9	The system should be able to understand most, if not all, commands given by the user, as long as they that fall under the systems recognized keywords
PNFR10	System should be able to support the modification of the visual settings, such as text size, button size, and having a simple and standard menu

[3] Issues with the Preliminary Definition Given

3.1.Domain Issues

Domain Issue ID	**Domain Issue Description**	
DI1	PD_ID	PD1. Indoor navigation complexity
	1. Each building has a different floor plan with varying levels of square footage, rooms, and floors.	
	Option 1	Work with schools/building managers to get a digital floorplan and upload the complete map of the building, so that the app can integrate it
	Option 2	Allow a caretaker to add and update map data
	Option 3	Sensors will alert user of walls and give basic directions that would mostly just prevent user from running into anything
	Choice	Option 1
	Rationale	Option one provides the most complete domain knowledge of the listed options. It is the most simple and usable option for getting a building's schematic.
Satisfied by		

Domain Issue ID	**Domain Issue Description**	
DI2	PD_ID	PD2. Obstacle detection ambiguity
	1. The project specification mentions detecting obstacles, but it is unclear what types of objects the app should detect 2. Different obstacles may have priority for avoidance	
	Option 1	Define objects into different categories such as static obstacles, temporary obstacles, and dynamic obstacles. Static obstacles would include walls, etc., temporary obstacles would include chairs, etc., and dynamic obstacles would include people, etc. Depending on the object type, the app would alert the user of possible detours whether it's to take a different route or take

		some amount of steps to the side to avoid, or alert user to stop for a moment in the case of a moving object
	Option 2	Use the phone's camera to detect obstacles in the user's way in real time
	Option 3	Let user/caretaker customize object detection preference
	Choice	Option 1
	Rationale	Option one provides the most complete obstacle detection and avoidance. It prevents the user from running into any type of object in real-time.
Satisfied by		

3.2. Stakeholder Issues

Stakeholder Issue ID	Stakeholder Issue Description	
SI1	PS_ID	PS1. Blind or visually impaired people who need to navigate indoors
	1. Blind and visually impaired covers a large range of disabilities that the application will have to accommodate.	
	Option 1	Find the largest range of blind and visually impaired that we can cover with the smallest number of functional requirements.
	Option 2	Find the range of blind and visually impaired that struggles the most within the domain that we can cover with the smallest number of functional requirements.
	Option 3	Find and research levels of visual impairment to improve domain knowledge and find requirements that satisfy each level.
	Choice	Option 3
	Rationale	Option 3 allows a tailored experience for users and also segmentation of the problem for ease of tracking documentation.
Satisfied by		

Stakeholder Issue ID	Stakeholder Issue Description	
SI2	PS_ID	PS2. Caretaker, someone who aides the blind person in configuring the application
	2. Caretaker is ambiguous, there is a large range of competency and availability that changes caretaker to caretaker	
	Option 1	Find the largest range of caretakers that we can cover with the smallest number of functional requirements.
	Option 2	Find the range of caretakers that struggles the most within the domain that we can cover with the smallest number of functional requirements.
	Choice	Option 1
	Rationale	Option 1 allows for the largest range of caretakers to be satisfied. Configuration should be a simple process.

Stakeholder Issue ID	Stakeholder Issue Description	
SI3	PS_ID	PS3. Accessibility department staff who may upload indoor maps/layouts
	3. Indoor maps/layouts will most likely differ in format from place to place.	
	Option 1	Find the largest range of accessibility department staff and their layouts that we can cover with the smallest number of functional requirements.
	Option 2	Find the range of department staff with the most complex layouts within the domain that we can cover with the smallest number of functional requirements.
	Choice	Option 1

	Rationale	Option 1 allows for the largest range of accessibility department staff to be satisfied.

3.3. Functional Requirements Issues

FR Issue ID	**Description**	
FRI1	PFR_ID	PFR1. Accept a location from the user to navigate to using voice commands or another form of user interface.
	1. Voice recognition may misinterpret user input. 2. Relying only on one input method may reduce accessibility.	
	Option 1	Only accept locations to go to through voice commands
	Option 2	Allow voice commands are different input methods
	Choice	Option 2
	Rationale	Supporting multiple input methods improves reliability in noisy environments and accessibility for diverse users.
Satisfied by	FR1	

FR Issue ID	**Description**	
FRI2	PFR_ID	PFR2. User can confirm destinations that have been suggested or selected by the user.
	1. System-suggested destinations may be wrong. 2. Navigating without confirmation may lead to errors.	
	Option 1	Automatically assume suggested destination is correct.
	Option 2	Require explicit user confirmation before navigation begins.

	Choice	Option 2
	Rationale	Confirmation prevents the wrong navigation and ensures the user has full control.
Satisfied by	FR2	

FR Issue ID	**Description**	
FRI3	PFR_ID	PFR3. Figure out multiple routes to the users confirmed destination and inform them of the options as well as describe each route including time, distance, and potential obstacles.
	1. Too many route options may overwhelm users. 2. Too few details may reduce trust in the system. accessibility.	
	Option 1	Provide only the shortest route.
	Option 2	Provide 2-3 best routes with details about each one.
	Choice	Option 2
	Rationale	Multiple route options with descriptions give the user informed choices.
Satisfied by	FR3	

FR Issue ID	**Description**	
FRI4	PFR_ID	PFR4. While navigating inform the user of their distance to the next direction.
	1. Too frequent updates could annoy users. 2. Too few updates could cause confusion and missed turns.	
	Option 1	Provide only one notification close to the turn.
	Option 2	Provide notifications at intervals before the turn.
	Choice	Option 2

	Rationale	Interval-based updates keep the user oriented and prepare them without providing too much information.
Satisfied by	FR4	

FR Issue ID	**Description**	
FRI5	PFR_ID	PFR5. While navigating, when reaching a stop or turn inform the user and tell them where to go.
	1. Instructions given too late may cause missed turns. 2. Ambiguous wording could confuse users. ssibility.	
	Option 1	Give a single alert only at the moment of the turn.
	Option 2	Give both advance notice and a confirmation at the turn.
	Choice	Option 2
	Rationale	Early and final confirmation ensures clarity and reduces the chance of mistakes.
Satisfied by	FR5	

FR Issue ID	**Description**	
FRI6	PFR_ID	PFR6. Detect obstacles using sensors and tell user how to avoid them.
	1. Sensors may produce false information about the surroundings. 2. Generic alerts may not be useful without avoidance instructions.	
	Option 1	Alert only when major obstacles are detected.
	Option 2	Alert for all detected obstacles with avoidance directions.
	Choice	Option 2

	Rationale	Providing full detection with clear instructions maximizes user safety.
Satisfied by	FR6	

FR Issue ID	Description	
FRI7	PFR_ID	PFR7. Place emergency calls and messages when a fall occurs or the system is lost.
	1. False alarms may cause unnecessary calls. 2. Manual-only calling may delay critical help.	
	Option 1	Require user to trigger emergency calls manually.
	Option 2	Automate emergency calls/messages with a cancel option for false alarms.
	Choice	Option 2
	Rationale	Automation ensures timely help, while the cancel option prevents unnecessary contacts.
Satisfied by	FR7	

FR Issue ID	Description	
FRI8	PFR_ID	PFR8. Predict what the user's actions would be in advance using their location habits and schedule, suggesting their next destination and accepting the user's choice.
	1. Predictions may be inaccurate. 2. Too many suggestions could overwhelm the user.	
	Option 1	Predict one destination and auto-navigate.
	Option 2	Suggest possible destinations but require confirmation.
	Choice	Option 2

	Rationale	Predictions can save time, but requiring confirmation ensures accuracy and control.
Satisfied by	FR8	

FR Issue ID	**Description**	
FRI9	PFR_ID	PFR9. User can create new emergency contacts for the system to use, including setting priority for their contacts
	1. Too many contacts without priority may cause delays. 2. Limiting to one contact reduces redundancy.	
	Option 1	Allow only one emergency contact.
	Option 2	Allow multiple contacts with priority ranking.
	Choice	Option 2
	Rationale	Prioritized contacts ensure redundancy while keeping communication efficient.
Satisfied by	FR9	

FR Issue ID	**Description**	
FRI10	PFR_ID	PFR10. User can customize settings such as volume, route update frequency, and language.
	1. Too many settings may overwhelm some users. 2. Lack of customization may reduce accessibility. accessibility.	
	Option 1	Provide only basic customization
	Option 2	Provide full customization including volume, frequency of updates, and language.
	Choice	Option 2

	Rationale	Full customization improves accessibility for diverse needs; users can ignore advanced options if not needed.
Satisfied by	FR10	

3.4.Non-Functional Requirements(NFR) Issues

NFR Issues ID	Description	
NFRI1	PNFR_ID	PNFR1. The system shall be secure.
	1. What is the definition of security? 2. Security allows for Confidentiality, Integrity and Authentication 3. If the users' information is not secure it could allow for unauthorized access to their data 4. Users might lose trust in and stop using the app altogether if they feel their data is not protected	
	Option1	Confidentiality
	Option2	Integrity
	Option3	Authentication
	Choice	1
	Rationale	For the particular smartphone app considered, confidentiality seems to be the most relevant (in relation to HIPAA).
Satisfied by	NFR1	

NFR Issues ID	Description	
NFRI2	PNFR_ID	PNFR2. The system shall be usable.
	1. What mechanism does the system support for Confidentiality?	

	2. Requiring extensive training or over complex authentication could result in less app users 3. If there is no security within the app users may not feel safe	
	Option1	Password
	Option2	Access Card
	Option3	Retinal Scan
	Choice	1 + 2
	Rationale	Easy to implement and cheap but also provides a satisfactory level of assurance.
Satisfied by	NFR2	

NFR Issues ID	**Description**	
NFRI3	PNFR_ID	PNFR3. Generating desired sentences and representing them pictorially as well as associating with a sound/voice.
	1. Solely relying on the voice prompts is not all inclusive 2. Not having any sort of visual representation might hinder some understanding	
	Option1	Only allowing for voice output
	Option2	Allowing for both voice outputs, as well as providing pictures as well
	Choice	2
	Rationale	Combining both video and audio will allow for a wider range of users to benefit from the app.
Satisfied by	NFR3	

NFR Issues ID	**Description**

NFRI4	PNFR_ID	PNFR4. The system has a response time of about 2-3 seconds for user input under normal operating commands/ conditions
	1. A delay from the system could cause the user to get further lost 2. Delays may force the users to navigate on their own, making the system unreliable	
	Option1	Optimize the system to allow for very minimal processing to reduce response time
	Option2	Use pre downloaded content to reduce delays, especially in high functioning areas
	Option3	Allow offline guidance if there is a delay
	Choice	1 + 2
	Rationale	The combination of minimal lightweight possessing with pre downloaded content will allow for a short processing time between 2-3 seconds
Satisfied by	NFR4	

NFR Issues ID	**Description**	
NFRI5	PNFR_ID	PNFR5. System should be available daily, allowing for multi-user access, ensuring especially high availability during navigation and emergency situation
	1. If the system is delayed during emergencies or delays emergency calls the users will see the app as being unreliable 2. If the system runs slowly when multiple accounts are accessing it then the app can also be seen as being unreliable as well	
	Option1	Implementing a cloud-based infrastructure, with failover information to ensure system stays

		robust even with multiple users, and during emergencies
	Option2	Having local fallback modes that enacts when the system seems to have service interruptions
	Option3	No implementation of fallback modes
	Choice	1+2
	Rationale	Combining these two options allows for a high operating system while multiple users are using the app, especially during emergencies
Satisfied by	NFR5	

NFR Issues ID	**Description**	
NFRI6	PNFR_ID	PNFR6. The system should securely save and encrypt users' personal data, such as their location history and username/email address. The system must also follow privacy protection rules and regulations
	1. If the user's information can be breeched, their data could be unlawfully exposed 2. Failure to follow the privacy protection laws can result in very intense legal action and lose user’s trust	
	Option1	Implement strong encryption when gathering and saving user information
	Option2	Set up access controls using authentication
	Option3	Store user data with little to no encryption to allow for faster access to information
	Choice	1+2

	Rationale	Implementing strong encryption and having a form of authentication for the user is necessary to gain user trust and keep information secure
Satisfied by	NFR6	
NFR Issues ID	**Description**	
NFRI7	PNFR_ID	PNFR7. The system should support the switching of multiple languages via settings, with clear voice communication and understanding of voice prompts for every language
	1. If only one or two languages are supported this will limit the amount of users 2. If the voice prompt cannot successfully read different languages this will also limit the amount of users	
	Option1	Implement the support of multiple languages using a known speech API tool
	Option2	Hardcode one or two languages within the system
	Choice	1
	Rationale	This allows for a large range of languages to be supported and used within the app. It is also proven to be highly supported and simpler to integrate then hardcoding multiple languages.
Satisfied by	NFR7	

NFR Issues ID	Description	
NFRI8	PNFR_ID	PNFR8. The system should be reliable in all areas whether there is low lighting, crowds, or loud noise conditions, should also support when the user reports new obstacles in real time

	1. It will become frustrating to users if they cannot add obstacles in real time, this leads the app to become outdated, more quickly 2. If the app is not reliable in low lighting or noisy areas this can cause for obstacles to be missed, and users may be harmed	
	Option1	Implement multiple sensors within the system, including voice isolation and low lighting enhancement
	Option2	Only rely on standard phone microphones and cameras
	Option3	Allow users to report different obstacles they find, both by voice and touch while also allowing for the system to remember these new obstacles
	Choice	1 + 3
	Rationale	The aspects described will allow the system performance to be reliable in all conditions while also allowing for obstacle updates which is very important to ensure maximum safety for the users’
Satisfied by	NFR8	

NFR Issues ID	**Description**	
NFRI9	PNFR_ID	PNFR9. The system should be able to understand most, if not all, commands given by the user if they fall under the systems recognized keywords
	1. If system fails to understand supported voice commands users will likely find this frustrating and can reduce their trust in the app	
	Option1	Allow for a flexible command recognition, could implement NLP (Natural Language Processing)
	Option2	Allow for strict wording restrictions by the users

	Choice	1
	Rationale	Allowing for flexible command recognition will make the app more user-friendly and accessible. Implementing NLP aspect of AI could be extremely innovative and help with the app become more adaptable long term.
Satisfied by	NFR9	

NFR Issues ID	**Description**	
NFRI10	PNFR_ID	PNFR10. System should be able to support the modification of the visual settings, such as text size, button size, and having a simple and standard menu
	1. If the menu is too complex some users may not be able to successfully navigate to their desired location 2. Overall accessibility will be reduced	
	Option1	Allow for users to adjust text size and button size, while also created a simple menu that can be easily navigated by all users
	Option2	Do not provide options for the user to adjust the text or button sizes, have an inconsistently drawn up menu
	Choice	1
	Rationale	This allows the app to be more accessible for a larger range of people, including those with disabilities
Satisfied by	NFR10	

[4] WRS

4.1.W

4.1.1. Problem

Problem ID	Problem Description	Corresponding Goals
P1	Voice-only input may be unreliable in noisy environments, reducing accessibility.	G1
P2	System-suggested destinations may be incorrect, leading to navigation errors.	G2
P3	Having too many options, or not enough options, can confuse or frustrate the user.	G3
P4	Navigation updates that are too frequent or too infrequent may confuse or frustrate the user.	G4
P5	Unspecific or late turn instruction can cause a user to miss their directions.	G5
P6	Detecting obstacles could lead to false positives or fail to detect an object.	G6
P7	Emergency detection needs to be automatic, while avoiding false alarms.	G7
P8	Predicting a destination could be inaccurate.	G8
P9	Contacting someone in the case of emergency assistance must have redundant systems while still ensuring a quick response.	G9
P10	Having a lot of complex settings could overwhelm the user but may be needed to provide the best accessibility experience.	G10
P11	User data and location information must remain secure in accordance with HIPPA laws.	G11

P12	The system should ensure a quick response time and high reliability while in use.	G12

4.1.2. Goals

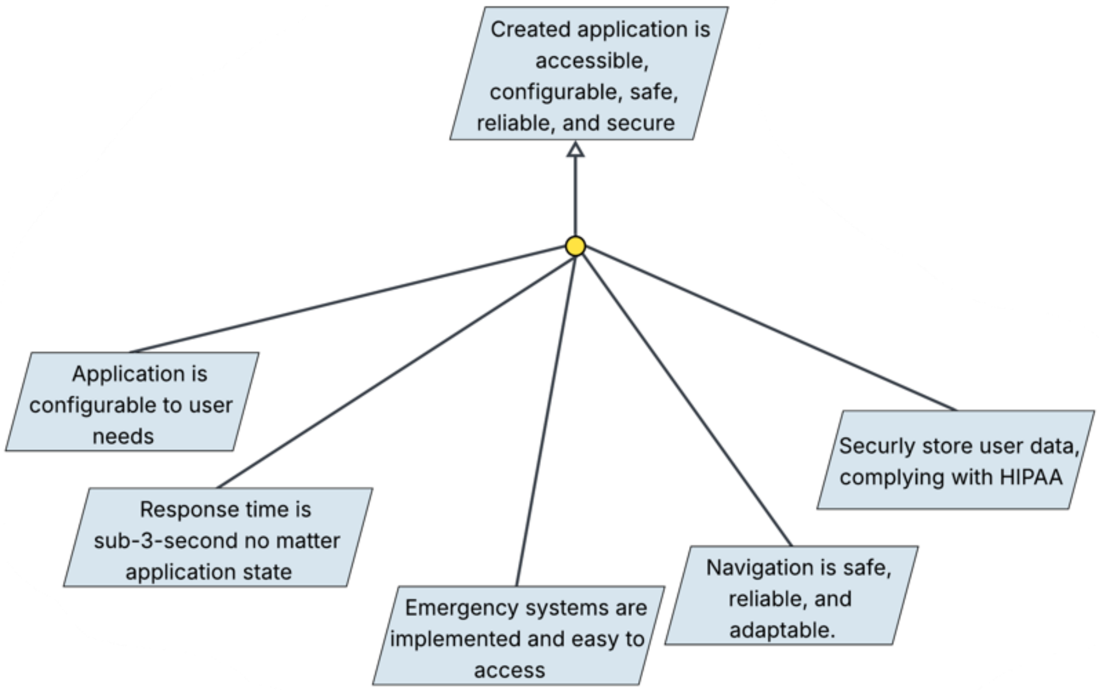


Figure 1. Goal: “Created application is accessible, configurable, safe, reliable, and secure”

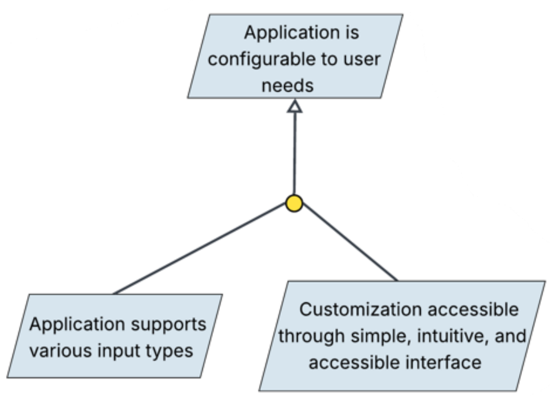


Figure 2. Goal: “Application is configurable to user needs”

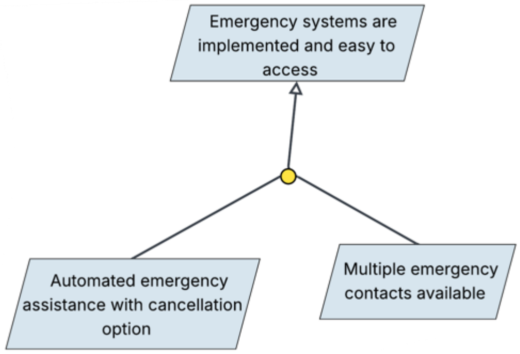


Figure 3. Goal: “Emergency systems are implemented and easy to access.”

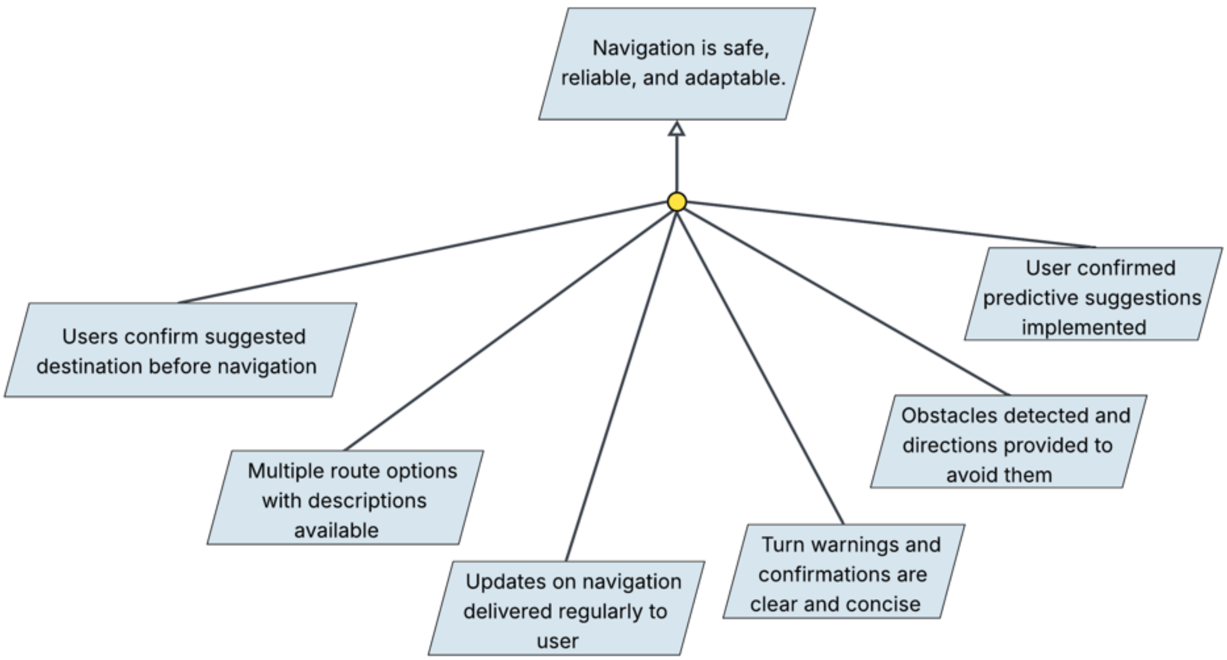


Figure 4. Goal: “Navigation is safe, reliable, and adaptable”

Goal ID	**Goal Description**	**Backward Traceability**	**Forward Traceability**
G1	Support for various input methods such as voice and touch to ensure the widest range in accessibly in any environment.	P1	IFRO1, INFRO3
G2	Users must confirm any suggest destination before navigation will start.	P2	IFRO2

G3	2-3 route options with descriptions of their time, distance, and possibly obstacles should be given.	P3	IFRO3
G4	Deliver navigation updates of a consistent interval to keep the users oriented.	P4	IFRO4
G5	Provide turn warnings before turning, along with a turn confirmation, all while using concise and clear wording.	P5	IFRO5
G6	Detect all obstacles and give concise direction to avoid them.	P6	IFRO6
G7	Automate emergency assistance with the ability to cancel it.	P7	IFRO7
G8	Offer predictive suggestion, but require user confirmation	P8	IFRO8
G9	Allow multiple emergency contacts, along with the ability to call emergency services immediately.	P9	IFRO9
G10	Provide application customization through a simple, intuitive, and accessible, interface.	P10	IFRO10
G11	Protect the user’s data by securely storing it and complying with HIPPA privacy laws.	P11	INFRO1, INFRO4
G12	Maintain reliability and at least a sub-3-second response time no matter the application state.	P12	INFRO2, INFRO4

4.1.3. Improved Understanding of Domain, Stakeholders, Functional, and Non-Functional Objectives

4.1.3.1. Improved Domain

Improved Domain ID	**Improved Domain Description**

ID1	The application operates within indoor environments that contain varying layouts, using building maps when possible and phone sensors to allow navigation for visually impaired users.
ID2	The application must process real-time environmental feedback to classify obstacles as static, temporary, or dynamic, to provide better, faster, safer, avoidance.

4.1.3.2. Stakeholders

- Primary stakeholder:
  - Visually impaired users, they depend on the app for safe indoor navigation
- Secondary stakeholders:
  - Caretakers, they configure the app settings and assist the user during an emergency
  - Emergency responders, they can be alerted by the app in an emergency
  - Developers, they ensure the constant reliability and system security

4.1.3.3. Improved Functional Objectives

Based on the above information and our goals, the functional objectives are:

Improved FR Objective ID	Objective Description	Alleviates Problems	Achieves Goals
IFRO1	Support for various input methods such as voice and touch.	P1	G1
IFRO2	Require user confirmation for suggested destinations before navigation begins.	P2	G2
IFRO3	Provide 2-3 optimized routes per destination, with details on distance, time, and obstacles.	P3	G3
IFRO4	Provide distance and direction updates at specific intervals before turns.	P4	G4
IFRO5	Issue before turn and at turn confirmations using clear instructions.	P5	G5
IFRO6	Detect and classify all obstacles and issue instructions on how to avoid them in real-time.	P6	G6

IFRO7	Automate emergency assistance with the ability to cancel it for any false alarm.	P7	G7
IFRO8	Predict destinations by analyzing user habits and require confirmation before routing.	P8	G8
IFRO9	Allow users to create multiple emergency contacts.	P9	G9
IFRO10	Provide an interface for adjusting settings that is accessible by various means such as voice or touch inputs.	P10	G10

4.1.3.4. Improved Non-Functional Objectives

Improved NFR Objective ID	Objective Description	Alleviates Problem	Achieves Goal
INFRO1	Ensure all user data and location data is in compliance with HIPPA privacy laws.	P11	G11
INFRO2	Guarantee a sub-3-second response time by utilizing techniques such as caching, pre-downloaded maps, and optimizing processing when possible.	P12	G12
INFRO3	Use a combination of audio and visual cues to increase accessibility for various impairment levels.	P1	G1
INFRO4	Maintain constant application functionality through cloud-based infrastructure, with local systems as a fallback, all while maintaining HIPPA compliance.	P12	G12
INFRO5	Protect the user’s application by providing real-time updates and system protection.	P11	G11

4.2.RS

4.2.1. Functional Requirements

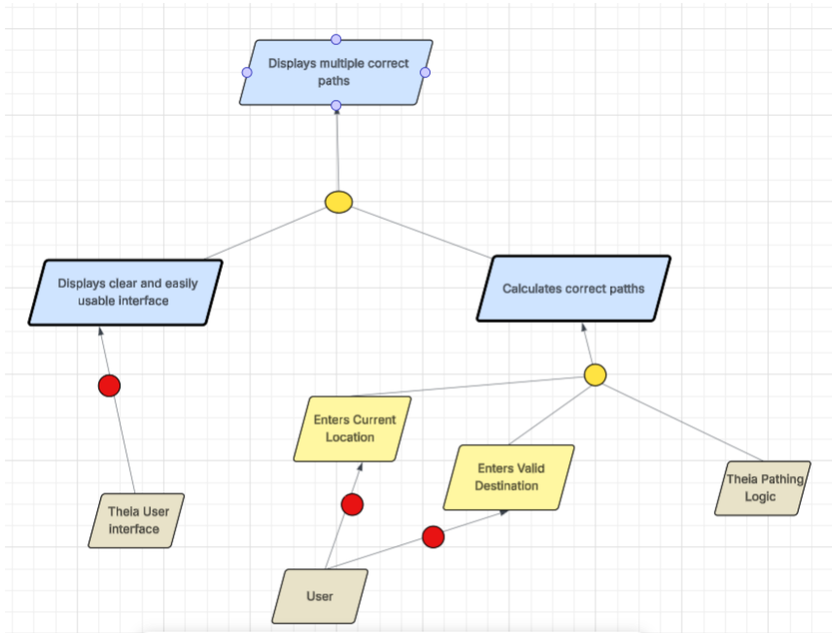


Figure 5. Responsibility: “Displaying multiple correct paths”

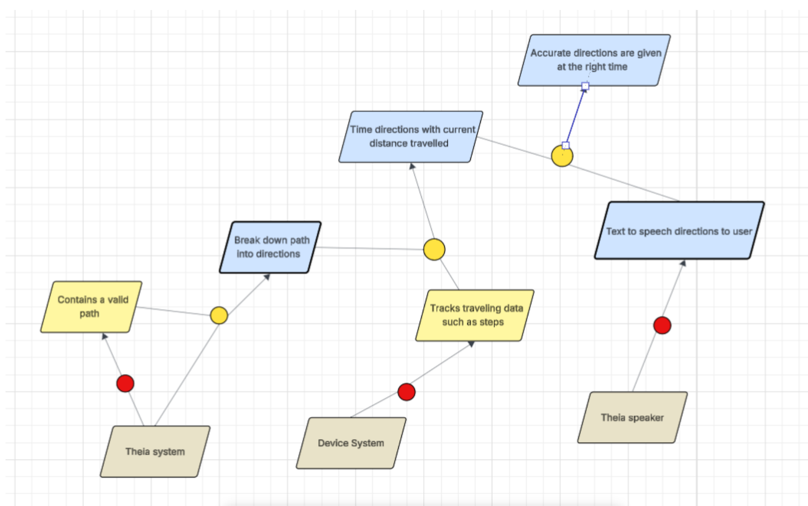


Figure 6. Responsibility: “Directions given are accurate”

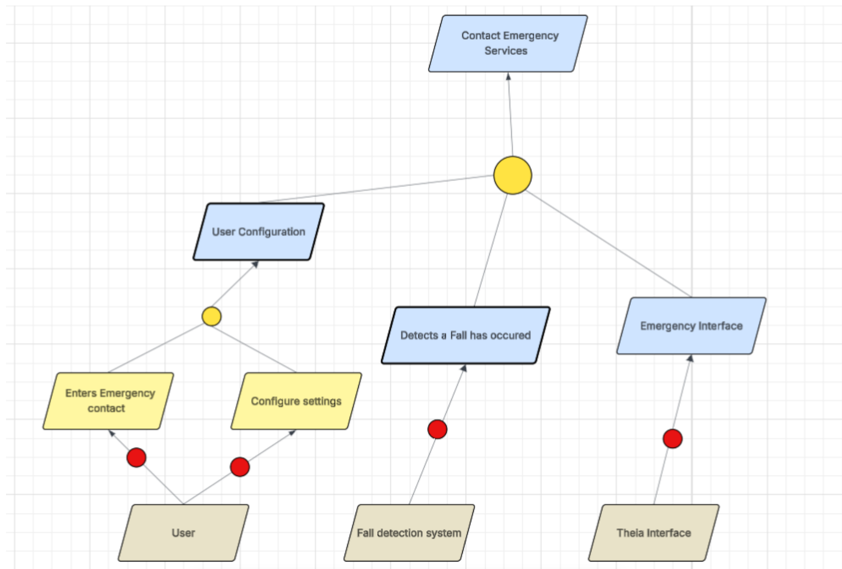


Figure 6. Responsibility: Contacting Emergency Services

FR ID	**Description**
FR1	Accept a location from the user to navigate to using voice commands or another form of user interface. This will be a tap interface which would allow users to select locations rather than text entry.
Satisfies Functional Requirement Issue	FRI1
Satisfies Objectives	IFR02, IFR03, IFR08
Satisfied by prototype feature	Address/Location entry
FR ID	**Description**
FR2	Users can confirm destinations that have been suggested or selected by the user.
Satisfies Functional Requirement Issue	FRI2
Satisfies Objectives	IFR02, IFR03, IFR08
Satisfied by prototype feature	Address/Location Entry
FR ID	**Description**
FR3	Figure out multiple routes to the users confirmed destination and inform them of the options as well as

	describe each route including time, distance, and potential obstacles.
Satisfies Functional Requirement Issue	FRI3
Satisfies Objectives	IFR03
Satisfied by prototype feature	Direction Suggestion
FR ID	**Description**
FR4	While navigating, inform the user of their distance to the next direction.
Satisfies Functional Requirement Issue	FRI4
Satisfies Objectives	IFR04, IFR05
Satisfied by prototype feature	Direction Updates during travel
FR ID	**Description**
FR5	When navigating, when reaching a stop or turn, inform the user and tell them where to go.
Satisfies Functional Requirement Issue	FRI5
Satisfies Objectives	IFR05
Satisfied by prototype feature	Turning updates during travel
FR ID	**Description**
FR6	Detect obstacles using sensors and tell the user how to avoid them.
Satisfies Functional Requirement Issue	FRI6
Satisfies Objectives	IFR06
Satisfied by prototype feature	Obstacle Detection
FR ID	**Description**
FR7	Place emergency calls and messages when a fall occurs, or the system is lost.

Satisfies Functional Requirement Issue	FRI7
Satisfies Objectives	IFR07
Satisfied by prototype feature	Emergency Contact
FR ID	**Description**
FR8	Predict what the user's actions would be in advance using their location habits and schedule, suggesting their next destination and accepting the user's choice.
Satisfies Functional Requirement Issue	FRI8
Satisfies Objectives	IFR08
Satisfied by prototype feature	Habit/prediction detection
FR ID	**Description**
FR9	User can create new emergency contacts for the system to use, including setting priority for their contacts
Satisfies Functional Requirement Issue	FRI9
Satisfies Objectives	IFR09
Satisfied by prototype feature	Emergency Contact editing
FR ID	**Description**
FR10	User can customize settings such as volume, route update frequency, and language.
Satisfies Functional Requirement Issue	FRI10
Satisfies Objectives	IFR10
Satisfied by prototype feature	User Settings

4.2.2. Non-Functional Requirements

NFR ID	Nonfunctional Requirement 1

NFR1	The system shall be secure, which means confidentiality.	
Operationalized Functional Requirements	OFR1	User data should only be viewable by appropriate staff and follow standards of privacy and security.
Satisfies Nonfunctional Requirement Issue	NFRI1	
Satisfies Non-functional Objective	NFO1	
Constrains	INF01	
Satisfied by prototype feature	Emergency Contact/User database	
NFR ID	**Nonfunctional Requirement 2**	
NFR2	The system shall be usable.	
Operationalized Functional Requirements	OFR1 OFR2	The system should be usable without proper vision with app user interface System can operate without user
Satisfies Nonfunctional Requirement Issue	NFRI2	
Satisfies Non-functional Objective	NFR02	
Constrains	INFR03	
Satisfied by prototype feature	System UI	
NFR ID	**Nonfunctional Requirement 3**	
NFR3	Generating desired sentences and representing them pictorially as well as associating with a sound/voice.	
Operationalized Functional Requirements	OFR1	Proper interpretation to user input
Satisfies Nonfunctional Requirement Issue	NFRI3	
Satisfies Non-functional Objective	NFO3	
Constrains	NFO3	
Satisfied by prototype feature	User Voice Recognition/Input	

NFR ID	**Nonfunctional Requirement 4**	
NFR4	The system has a response time of about 2-3 seconds for user input under normal operating commands/ conditions	
Operationalized Functional Requirements	OFR1	All functions must resolve by the 3rd second after prompting
Satisfies Nonfunctional Requirement Issue	NFRI4	
Satisfies Non-functional Objective	NFO2	
Constrains	INFR02	
Satisfied by prototype feature	App optimization	
NFR ID	**Nonfunctional Requirement 5**	
NFR5	System should be available daily, allowing for multi-user access, ensuring especially high availability during navigation and emergency situation	
Operationalized Functional Requirements	OFR1	Maintain operations during all times.
Satisfies Nonfunctional Requirement Issue	NFRI5	
Satisfies Non-functional Objective	INF04	
Constrains	INF04	
Satisfied by prototype feature	Cloud based system	
NFR ID	**Nonfunctional Requirement 6**	
NFR6	The system should securely save and encrypt users' personal data, such as their location history and username/email address. The system must also follow privacy protection rules and regulations	
Operationalized Functional Requirements	OFR1	Data should be properly saved and secured from unauthorized third parties.
Satisfies Nonfunctional Requirement Issue	NFRI6	

Satisfies Non-functional Objective	INFO1	
Constrains	INF01	
Satisfied by prototype feature	User Database	
NFR ID	**Nonfunctional Requirement 7**	
NFR7	The system should support the switching of multiple languages via settings, with clear voice communication and understanding of voice prompts for every language	
Operationalized Functional Requirements	OFR1	There should exist a language selection in user customization
Satisfies Nonfunctional Requirement Issue	NFRI7	
Satisfies Non-functional Objective	INF02, INF03	
Constrains	INF03	
Satisfied by prototype feature	User Settings	
NFR ID	**Nonfunctional Requirement 8**	
NFR8	The system should be reliable in all areas whether there is low lighting, crowds, or loud noise conditions, should also support when the user reports new obstacles in real time	
Operationalized Functional Requirements	OFR1	Should be reliable in various environments in obstacle detection
Satisfies Nonfunctional Requirement Issue	NFRI8	
Satisfies Non-functional Objective	INF02	
Constrains	INF02	
Satisfied by prototype feature	Obstacle Detection	
NFR ID	**Nonfunctional Requirement 9**	
NFR9	The system should be able to understand most, if not all, commands given by the user as long as they that fall under the systems recognized keywords	

Operationalized Functional Requirements	OFR1	Should recognize users regardless of their speech patterns if they follow general guidelines/user manual
Satisfies Nonfunctional Requirement Issue	NFRI9	
Satisfies Non-functional Objective	INFO3, INFO4	
Constrains	INFO4	
Satisfied by prototype feature	User Voice Input	
NFR ID	**Nonfunctional Requirement 10**	
NFR10	System should be able to support the modification of the visual settings, such as text size, button size, and having a simple and standard menu	
Operationalized Functional Requirements	OFR1	You should be able to manipulate user interfaces to personal preferences.
Satisfies Nonfunctional Requirement Issue	NFRI10	
Satisfies Non-functional Objective	INF03	
Constrains	INF03	
Satisfied by prototype feature	User Settings	

4.2.3. Specifications

Functional Specification ID	**Functional Requirement**
FS1	The system will provide at least 2 routes to your destination while providing details such as distance in steps or feet, the estimated time of travel in minutes, and should take no longer than 5 seconds to generate.
Satisfies Functional Requirement	FR3
Satisfies Objectives	IFR3
Satisfied by prototype feature	Route Suggestion
Functional Specification ID	**Functional Requirement**

FS2	The system should have around an 80 percent accuracy when parsing and interpreting voice input for current location and destination for travel.
Satisfies Functional Requirement	IFR1, IFR2
Satisfies Objectives	IFR01, IFR02, IFR08
Satisfied by prototype feature	Voice Input
Functional Specification ID	**Functional Requirement**
FS3	The system shall notify the user 10 meters before a turn and confirm the turn during the action using voice and optional vibration feedback. The distances will be tracked by the device the system is running on likely in the form of meters.
Satisfies Functional Requirement	FR4, FR5
Satisfies Objectives	IFR04, IFR05
Satisfied by prototype feature	Direction Notifications
Functional Specification ID	**Functional Requirement**
FS4	The system shall detect static and moving obstacles using onboard sensors and instruct the user to stop or change direction accordingly in under 2 seconds. This will rely on a camera being connected to the device allowing Theia to process the environment and recognize obstacles
Satisfies Functional Requirement	FR6
Satisfies Objectives	IFR06, INF02, INF03
Satisfied by prototype feature	Obstacle Detection
Functional Specification ID	**Functional Requirement**
FS5	If a sudden stop or fall is detected by the system, it shall initiate an emergency alert, notify all configured emergency contacts, and provide a 5-second cancellation option. The fall or stop should involve some device implementation that would allow us to

	detect sudden stops or falls. The emergency contact will also follow our design for configuration of emergency contacts in the settings.
Satisfies Functional Requirement	FR7
Satisfies Objectives	IFR07, INF01
Satisfied by prototype feature	Emergency Fall Detection
Functional Specification ID	**Functional Requirement**
FS6	The system shall analyze the last 5 navigation history logs to suggest the most likely next destination at startup. These can be stored in our database for each user, which upon relaunching we will provide their most frequently visited location as a selectable option which would immediately start navigation.
Satisfies Functional Requirement	FR8
Satisfies Objectives	IFR8, NFR9
Satisfied by prototype feature	Habit Detection/History of navigation
Functional Specification ID	**Functional Requirement**
FS7	The user shall be able to add or remove emergency contacts through a voice or touch interface and prioritize each contact. We will provide entries for a new contact, and users or any caretakers can setup the information necessary for a contact. Additionally, the configuration will provide a list of contacts which users will be able to highlight prioritize or remove from their contacts. 911 or state emergency services will be defaulted.
Satisfies Functional Requirement	FR9
Satisfies Objectives	IFR09, INF03
Satisfied by prototype feature	Emergency Contact Customization
Functional Specification ID	**Functional Requirement**

FS8	In the settings menu, the user shall be able to adjust voice volume, interface language, route update frequency, and enable/disable visual feedback. Our user interface should keep track of these values and update each page to respond to any changes in the settings.
Satisfies Functional Requirement	FR10
Satisfies Objectives	IFR10, NFR03, NFR10
Satisfied by prototype feature	User Customization
Functional Specification ID	**Functional Requirement**
FS9	All user-entered personal information, including email, name, and location history, shall be stored using some kind of encryption and accessed only with proper authentication. This database will be held by developers and proper channels to ensure the highest means of security for every user. This should also be reliable system as these key systems should have minimal downtime.
Satisfies Functional Requirement	NFR1
Satisfies Objectives	INFR01, INFR05
Satisfied by prototype feature	User Database

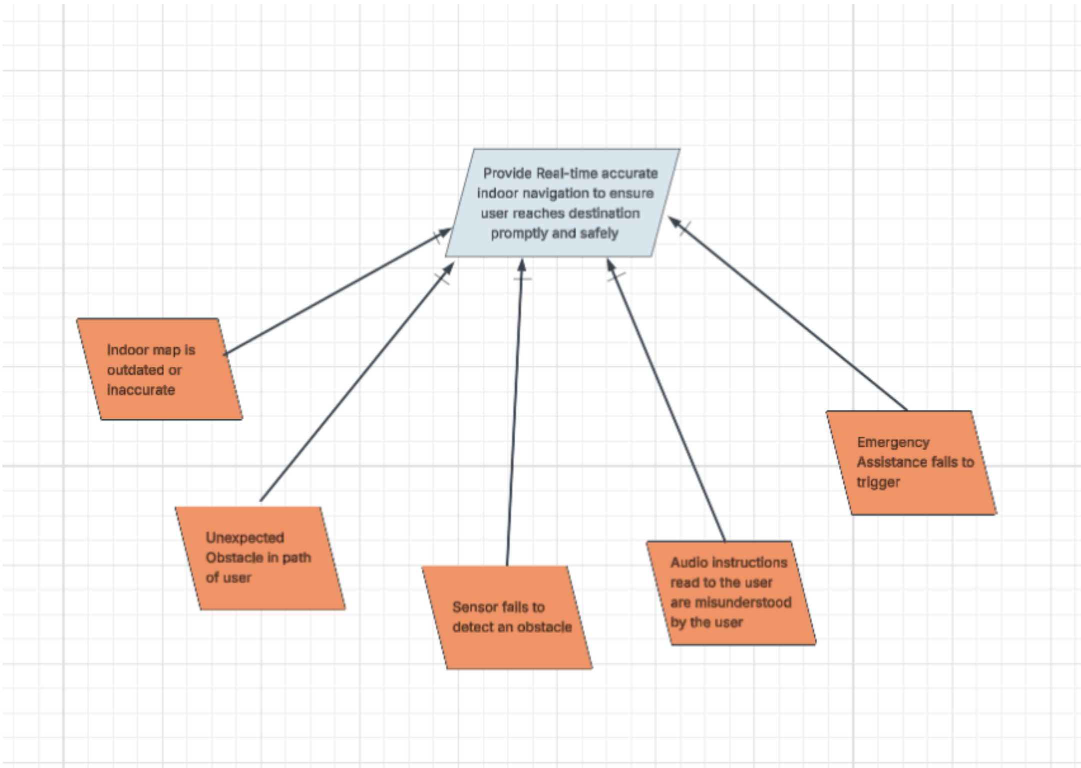
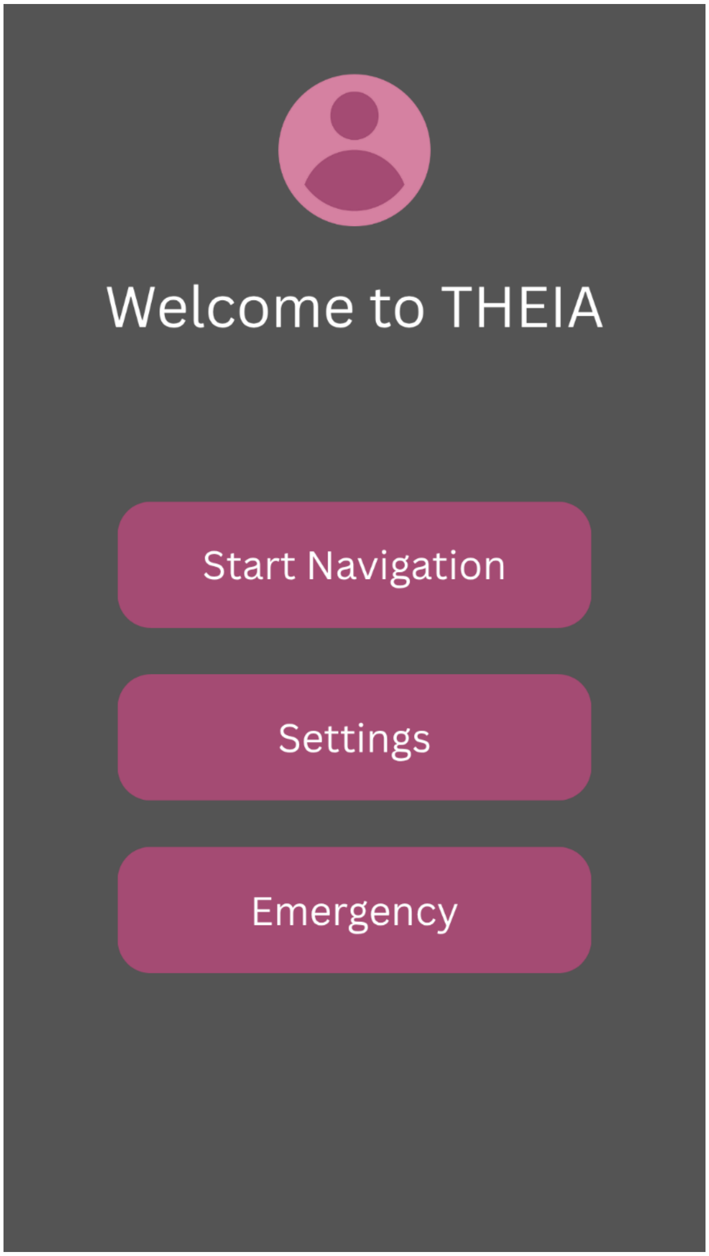


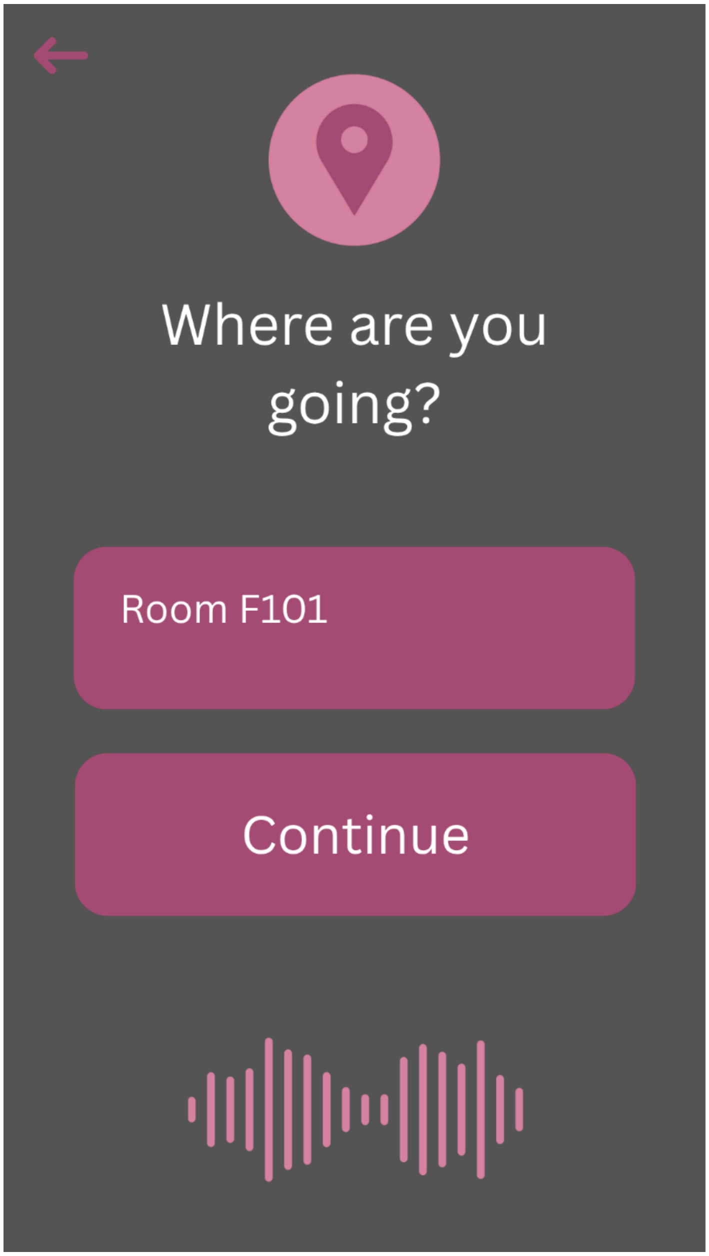
Figure 6. Obstacle Resolution: “Safe Navigation”

[5] Preliminary Prototype
Please view the Prototype Interface Mock-ups and the User manual.

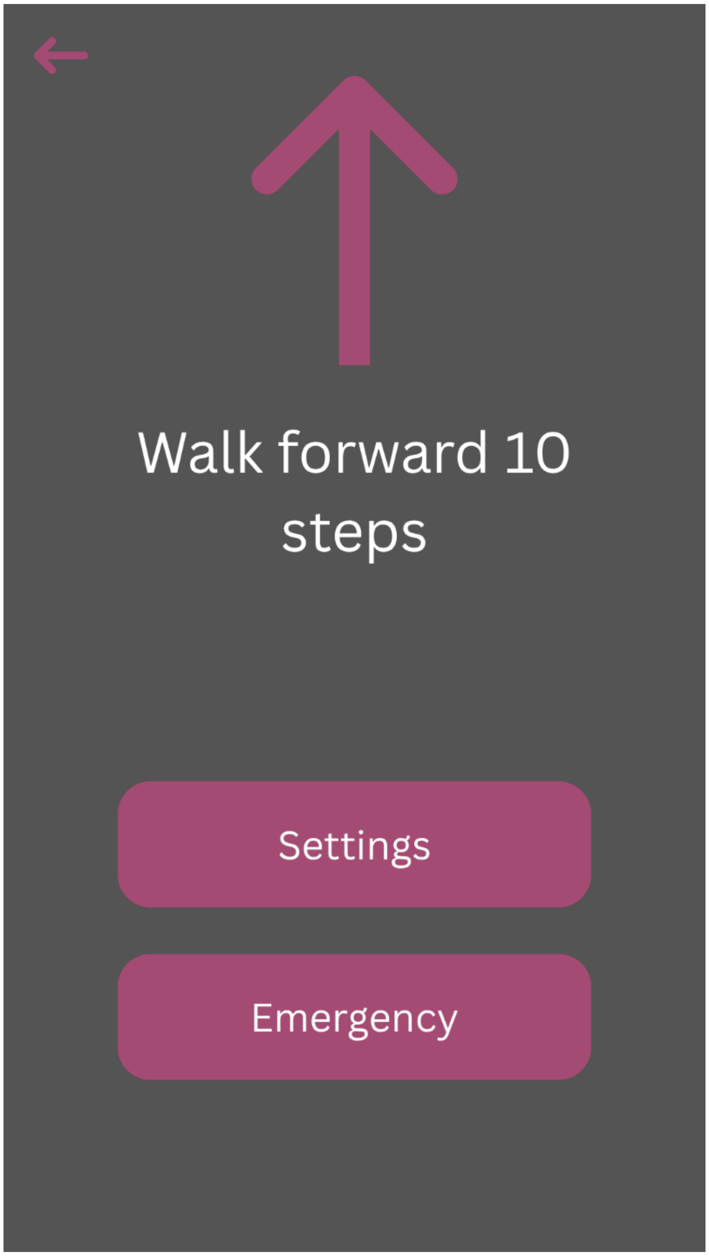
[6] Prototype Interface Mock-ups
THEIA Home



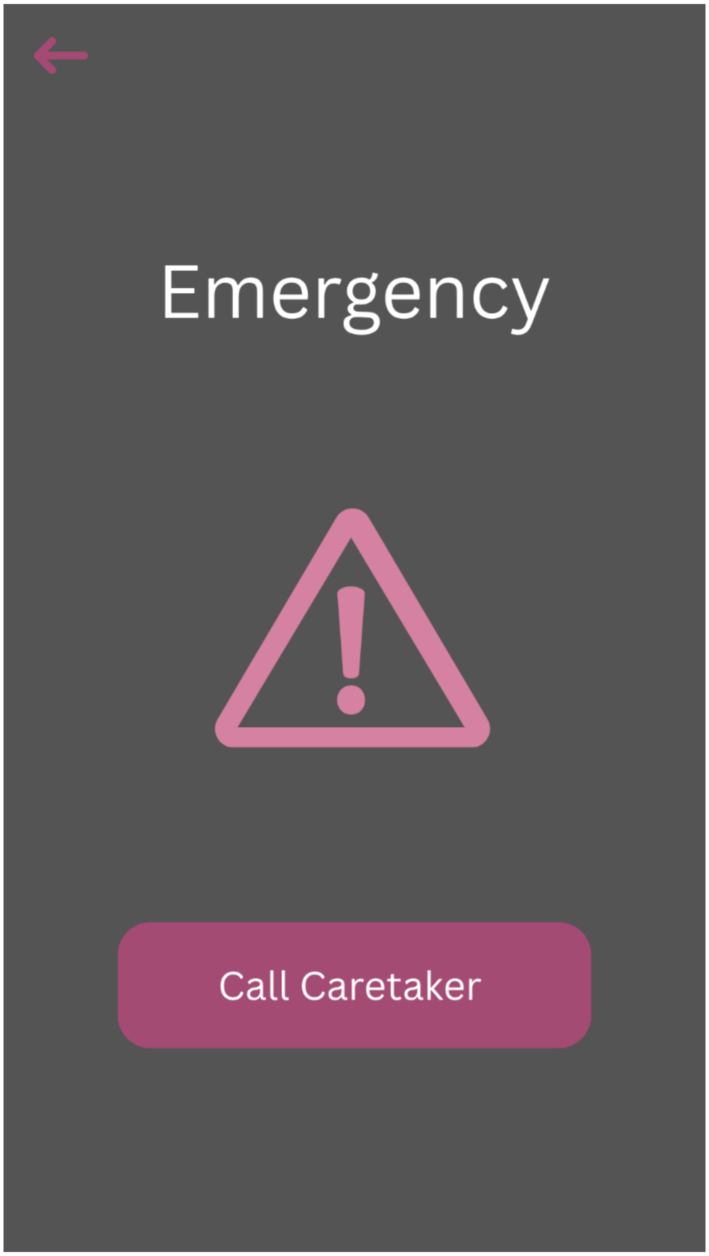
THEIA Input Navigation



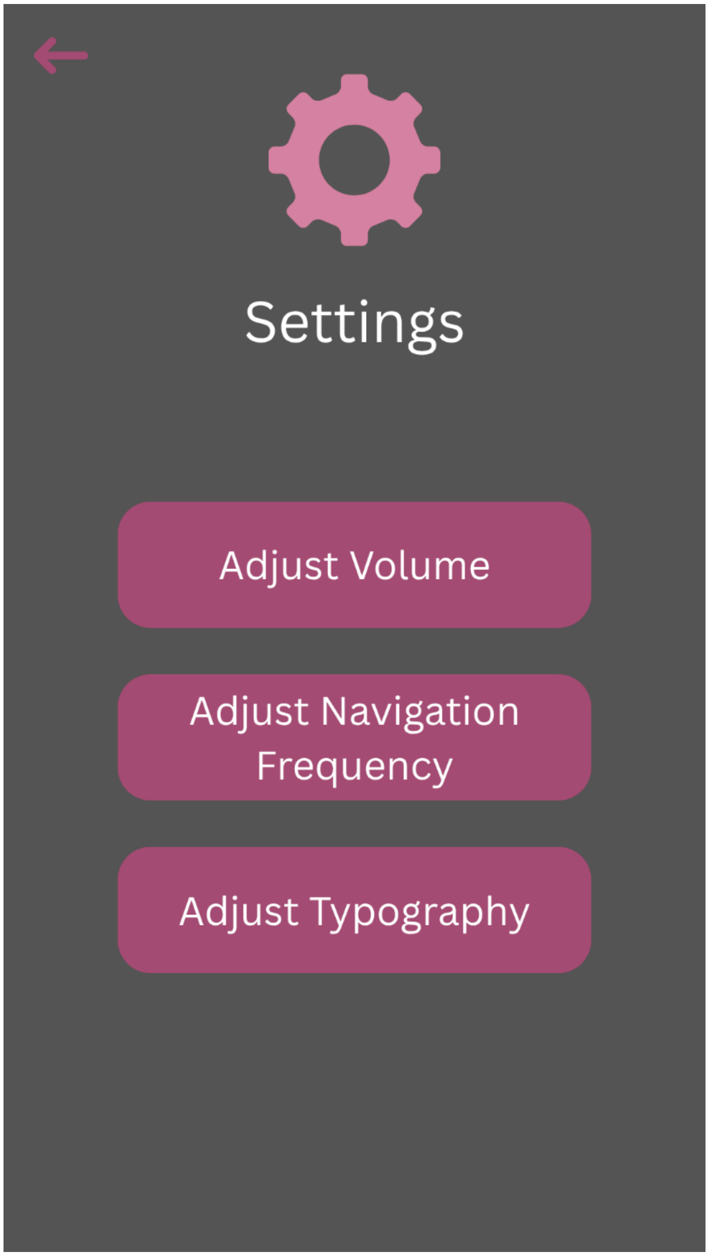
THEIA Navigation



THEIA Emergency



THEIA Settings



[7] User Manual

The home screen of the app will allow the user to select whether to go to navigation, settings, or emergency response. The navigation option will prompt the user for their next location and allow the user to either type the location or speak it. The app will then show the number of steps until they have to turn, as well as speak it. Once the user has made it to their destination, the app will return to the home screen. The settings option allows the user or caretaker to tailor the app towards the user. This includes increased vocal navigation for those who need it. The emergency option will ask the user if they are ok, and if there is no response within 15 seconds, the caretaker will automatically be called.

[8] Traceability

Functional

Functional Requirement	Functional Objective	Goals	Problem	Stakeholder
PFR1	IFRO1	G1, G11	P1, P11	Visually impaired user
PFR2	IFRO2, IFRO8	G2, G8, G12	P2, G8, P12	Visually impaired user
PFR3	IFRO3	G1, G3	P1, P3	Visually impaired user
PFR4	IFRO4	G4, G11, G12	P4, P11, P12	Visually impaired user
PFR5	IFRO5	G5	P5	Visually impaired user
PFR6	IFRO6	G6	P6	Visually impaired user
PFR7	IFRO7	G7	P7	Visually impaired user, Emergency responder, Caretaker
PFR8	IFRO8	G8	P8	Visually impaired user
PFR9	IFRO9	G9	P9	Visually impaired user, Caretaker
PFR10	IFRO10	G10	P10	Visually impaired user, Caretaker

Non-Functional

Non-functinonal Requirement	Non-Functional Objective	Goals	Problem	Stakeholder
PNFR1	INFRO1, INFRO4, INFRO5	G11, G12	P11, P12	Visually impaired user, Developer
PNFR2	INFRO3, INFRO4, INFRO5	G1, G11, G12	P1, P11, P12	Visually impaired user, Caretaker, Developer
PNFR3	INFRO2, INFRO3	G1, G12	P1, P12	Visually impaired user, Caretaker
PNFR4	INFRO2, INFRO5	G11, G12	P11, P12	Visually impaired user
PNFR5	INFRO4	G12	P12	Visually impaired user, Caretaker,

				Emergency responder, Developer
PNFR6	INFRO1, INFRO5	G11	P11	Visually impaired user, Developer
PNFR7	INFRO3	G1	P1	Visually impaired user
PNFR8	INFRO2, INFRO4	G12	P12	Visually impaired user
PNFR9	INFRO3	G1	P1	Visually impaired user
PNFR10	INFRO3	G1	P1	Visually impaired user, Caretaker, Developer

[9] Appendix I: Process Details

9.1.1. Meetings

Meeting Date	Topic of Discussion
9/3/2025	Introduction call, devised roles for every team member, mutually created a set of rules that everyone will follow, started working on the prelim plan document
9/10/2025	Checkup on Prelim plan and everyone's status on assigned work, discussed any questions/ concerns that arose while working on the document
9/17/2025	Discussed the plan of action for the WRS document, went over slides/ video from class to ensure complete understanding
9/23/2025	Discussed problems detailing the Prelim plan with the professor, met shortly after as a group to discuss updates on the WRS and any questions or concerns
10/7/2025	Discussed the video slides for the presentation, went over remaining work that needed to be completed and revised for the WRS document, updated completion dates

[10]References

We do not currently have any references.